



Product Assignment Part 2

Innovative solutions that simplify teaching experience of UG Faculty



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01 Key Challenges [03-03]

Understanding top challenges of American college classrooms that hamper the teaching experience in class

02 Language Barrier Issue [04-05]

Understanding and solving for the language barrier issue for non-native English speaking students in US colleges

03 Faculty Student Interaction [06-11]

Addressing the faculty student interaction issue during live classes and providing a solution that fixes the same

04 User Flows, Metrics and Next [12-15]

Main user flows of the product solution; What metrics to track? And the next steps for this solution

Understanding top challenges of American college classrooms

The problems that students in US colleges face can be categorised in the following buckets –

- Problems faced outside of the classroom (Managing your finances, time management, mental health issues)
- Problems faced during the class (as mentioned below)

Apart from Internet research, I interviewed 3 of friends and seniors studying in US to understand their perspective and problems. For the purpose of the assignment, we are focusing on the problems faced by students during their classes. Solving for in-class issues that students and faculties face will help in simplifying teaching experience for faculties.



Engagement in Class

50% students say they do not feel engaged in what they are learning in the class and around 60% students never ask a question in the classroom!

Issue 1



Language Barrier

Even if international and non-native English speaking students are proficient in English, they may struggle with accents, slang, and idiomatic expressions that are unfamiliar to them.

Issue 2



ROI of Learning

Students are unable to correlate the knowledge in the classroom to what is expected of them on the job. A direct reference or a link to their career options is missing from class.

Issue 3

Language barrier issue in American classrooms

What is the language barrier issue in American classrooms?

A major challenge for international students and non-native English speaking students in the USA is the language barrier. Even if students are proficient in English, they may struggle with accents, slang, and idiomatic expressions that are unfamiliar to them.

01

English as a second language

Currently over 20% of college students in USA do not speak English at home i.e. English is their second language

02

Demographic of US Colleges

18% of the college going students in USA in 2020 were non-native English speakers and in the same year there were 5% international students in US

03

Problem in understanding during class

Non-English speaking students have to refer to study material outside class to understand coursework. During class they google words to understand terms.

04

Accents and Dialects

There are around 150 different English dialects around the world out of which 30 major ones are spoken in America alone.

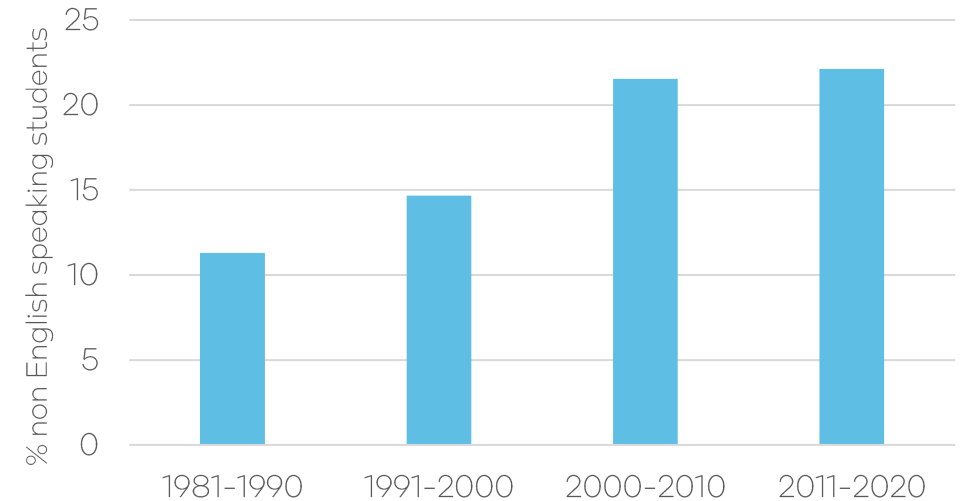
05

Standardized Test have their limitations

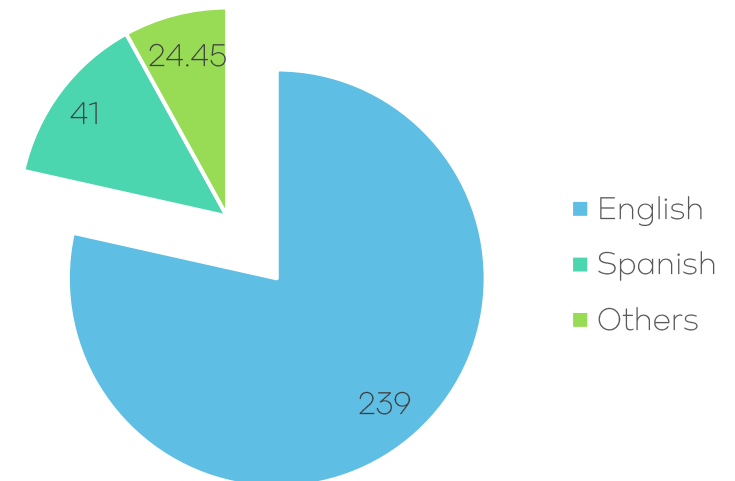
Tests like TOEFL, IELTS, GRE only show how well one performs in the English language but does not ensure that students will understand everything in class.

One approach would be to try and 'improve' English for non native speakers, but that wouldn't address issues related to all the factors mentioned above. Instead we can look at issue from the 'demand' side...

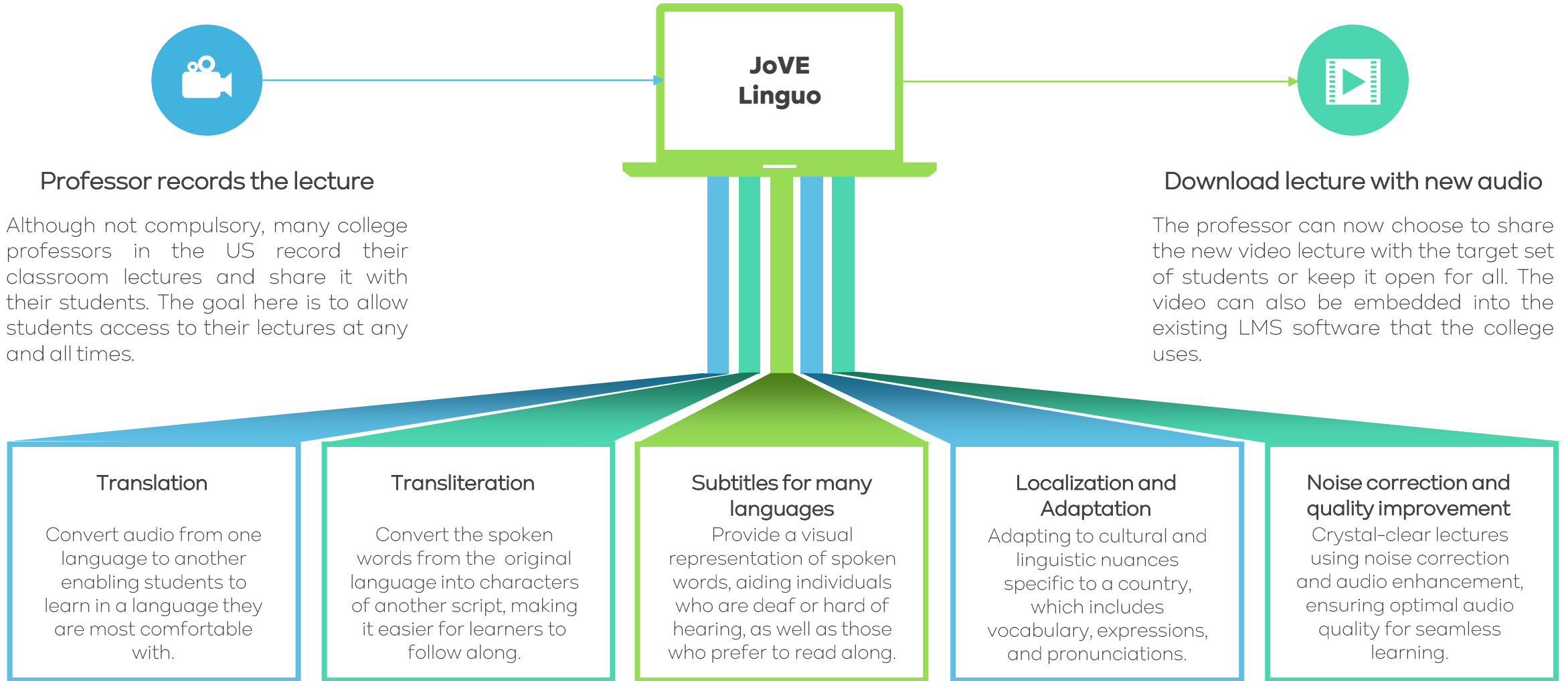
Percentage of school children who speak another language than English at home in the U.S. from 1979 to 2019



Languages of the US Population



JoVE Linguo – Translation, Subtitles and more!



There are many advanced tools like Amara, Descript, Trint (Paid tools) as well as Audacity, Podcastle (Free tools) which can be plugged and used for JoVE Linguo. Additionally a subject matter expert can be involved to improve the training of tools towards classroom lecture videos' audio translation.

Issue of Engagement in American College Classrooms

Defining engagement in a classroom

Prompting student participation by giving students the opportunity to ask or answer questions in front of the whole class is a common way for instructors to engage students in the college science classroom. Sometimes dubbed the “Socratic method” of teaching, studies have shown that encouraging students to ask and answer questions is frequently employed in both small-enrollment and large-enrollment college courses.

01

Numbers do not fit

[2021] 19 million students (2 yr & 4 yr) / 0.8 million full time faculty $\approx$ 24 and Avg. no. of students in a US College Classroom range from 80 to upto 200!

02

Students do not feel engaged in classroom

[2023 Gradient Learning Poll] 50% of the students say they are not engaged in what they are learning in school most of the time.

03

Student Faculty interactions are decreasing

[2019 NSSE report] Over 60% of the students rate their ‘Student-Faculty Interaction Score’ as ‘Medium’ or ‘Low’.

04

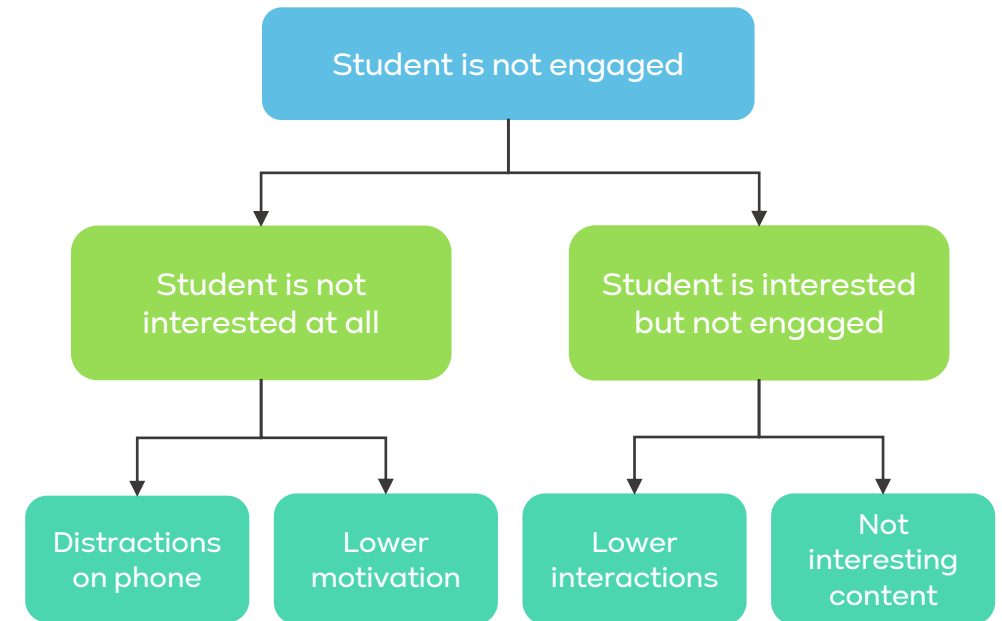
Engaged students are more likely to succeed

[Gallup Student Poll 2018] Engaged students are 2.5 times more likely to achieve excellent grades and do well in school

05

Student Questions in Classrooms

In a survey conducted for some colleges in southwest united states, over 63% of the students mentioned that they do not ask questions in class



5 Qs per class

Assuming an average class size of 80 students and an average class length of 1 hour. If the professor sets aside 15 mins for discussion. Effectively only 5 questions can be answered assuming 3 min per question-response cycle.

Why existing solutions aren't working?



Upvoting is Broken

'What is X', 'Did not understand X', 'Can you please explain X and Y again' all end up on the top of the priority due to 'Upvote' from students.

Checking Live Understanding of students

Imagine a 'LinkedIn Learning' like pop up question for students to answer. For a classroom presentation use case this is missing in all these tools.

Canvas Discussion and 'Office hours happen outside classroom

Canvas provides a forum for students to discuss questions, but this happens post class. By that time context is lost and students have forgotten their Qs.

Students aren't always ideal

Some students would always want to create nuisance. Professors in US have a 'classroom' policy but that doesn't stop such students.

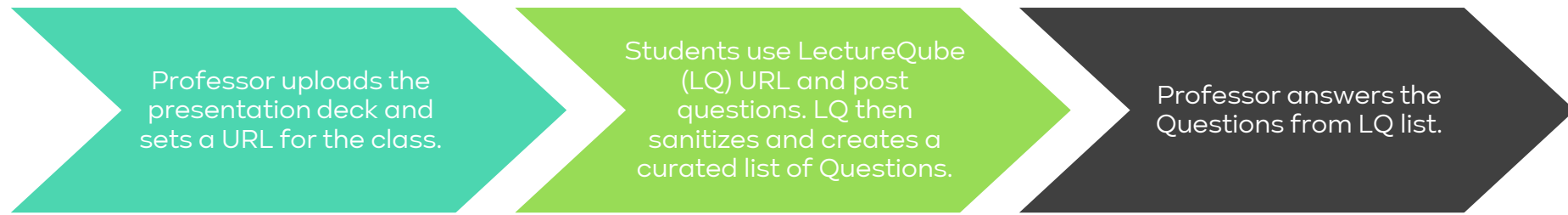
Comprehensive post class report missing

Once the class is over, the use of the tool is also considered complete, but the accumulated data can provide a lot of insights!

No history or past data being maintained

Maintaining past data of questions during lectures can help professors improve their classroom material and ensure query resolution in lecture itself

JoVE LectureQube



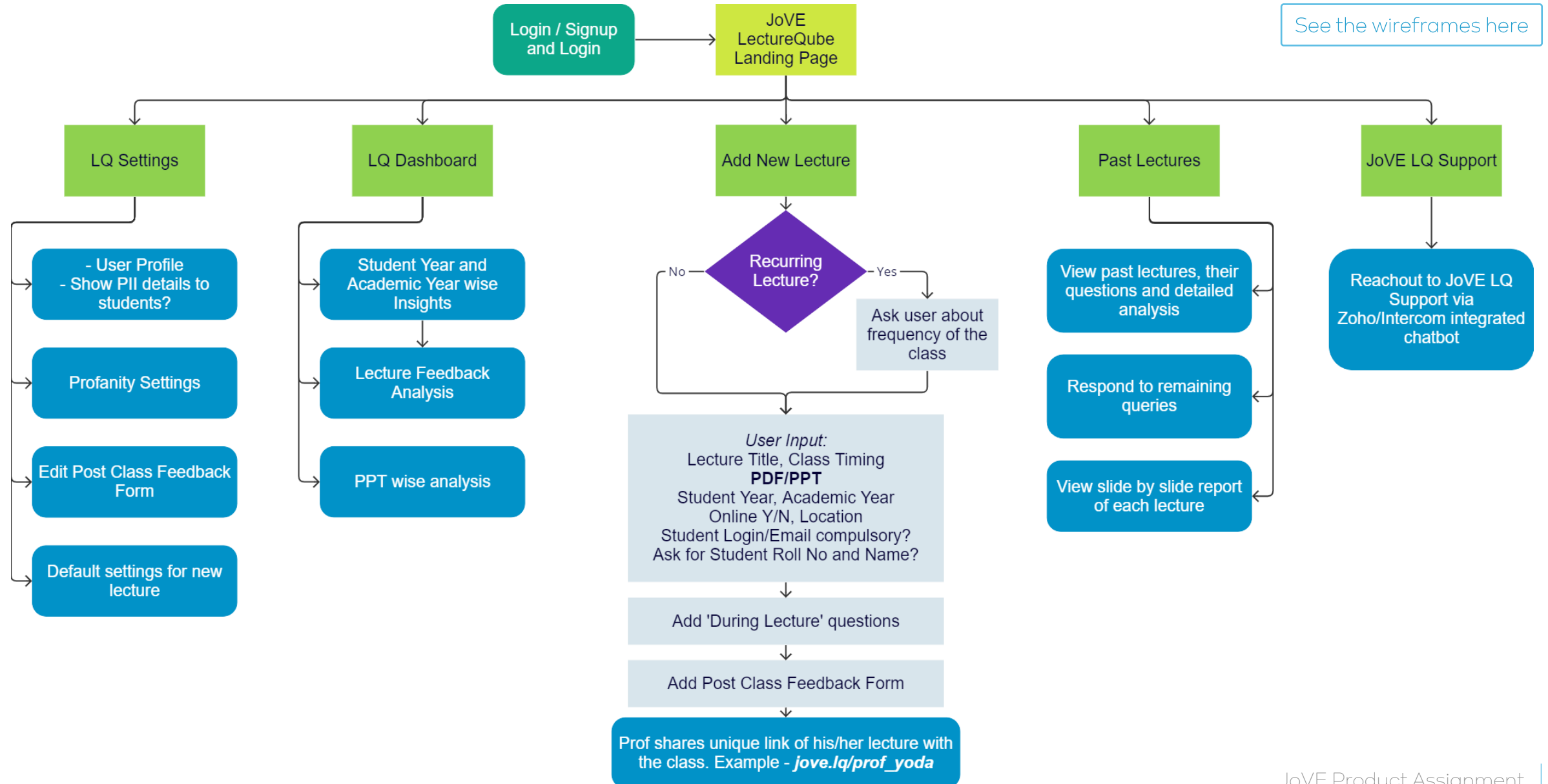
In an average class of 100 students, around 20 are expected to have one or more questions by the end of the class

Although not compulsory, most professor now prefer PPT mode of teaching in the classroom

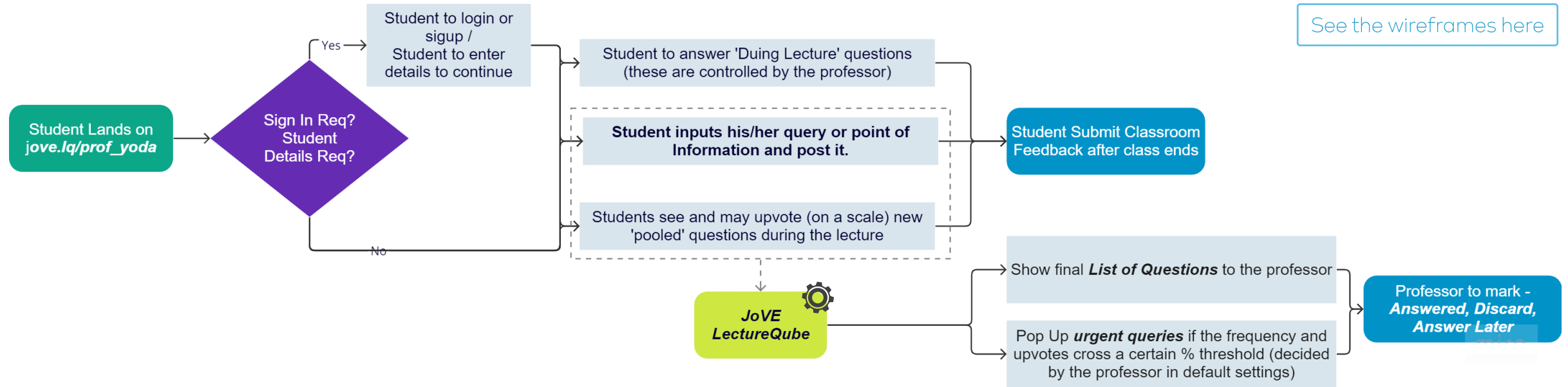
95% of adults in the age group 18-29 own a Smartphone in the United States

Two major categories of distractions because of smartphones in classes are – Texting and Social Media scrolling

JoVE LectureQube – Pre Class (Professor) (Information Arch.)



JoVE LectureQube – During the Class (Professor and Student)



AI and NLP tools like Google Cloud Natural Language, IBM Watson NLU, MS Azure Cognitive Services (all paid tools) can be integrated to achieve the capabilities of JoVE LQ. Free tools like NLTK and Stanford NLP can also be used, but would require in-house expertise to manage the same



Pooling Questions and Priority

A, A' and A'' pooled together as A

Decoupling Questions

A+B, A+C, A to be treated as A, B, C

Profanity Management

Filtering out inappropriate language

Advanced Upvoting

Voting on a scale only on pooled questions

Live Attention Score of Class

Monitoring the time spent on JoVE LQ

JoVE LectureQube – Post Class (Professor)



- Average time spent on LQ / Not spent on LQ
- Attention Score of the batch / class
- Common query topics in the class
- Topics where attention score was high

Student/Academic Year Wise Insights



- Feedback Score trend per class
- Course Material Feedback per class
- Custom questions included in the feedback
- Suggestions from students for the class

Lecture Feedback Analysis



- No. of questions per slide and their topic
- Average time spent on each slide
- Overall Lecture PPT feedback from students
- Suggestions from LQ on how to improve PPT

Lecture PPT Analysis



- Specific areas in the lecture with most Qs
- Common topics among unanswered Qs
- All student queries' raw data
- Send answer for remaining Qs to students

Past Year Classes

Edge Cases, Metrics and MVP



Edge Cases / Possible Pitfalls

- False Positives/Negatives in Question Sanitization
- Misinterpretation of Language (Dialect, slangs etc.)
- JoVE PPT feedback subjectivity
- Bias in question prioritization
- Sensitive topics identification



Metrics to track for JoVE LQ

- *North Star* – Avg. No. of questions per active student
- Avg. reuploads per PPT post classroom feedback
- Avg. no. of pop quizzes rolled out per active professor
- Avg. no. answered questions per active professor
- Avg. session time per active student



Things to consider for MVP

- For the MVP, we must focus on the core offering of JoVE LectureQube, which is 'AI Powered Question Sanitization and Grouping. The course material enhancement features, feedback feature and pop quiz feature can be developed once we have a positive response for the MVP.

Cost Estimation and Pricing of the Product



Development – Design – Product – QA Cost

Cost = person-hours*salary per hour for each of the mentioned roles. Additionally if an AI expert is involved, that cost would also be included.

01

02

Operational costs and Maintenance cost

If a 3rd party tool is used for JoVE LQ, then a running cost will be incurred. Additionally, tech support would be maintenance cost.



Analyzing Competitors' Pricing

To understand 'affordability' of such products for our target customers, we can look at competitors with highest market share and their pricing.

03

04

Choosing a pricing model

Given that there is significant running cost for the product, a credit based pricing model should be used for JoVE LQ.



Go-to-market strategy for JoVE LectureQube

Launch Strategy

Collaborating with top faculty and student orgs to use JoVE LQ during their workshops and seminars and collecting feedback on the product.

User Testimonials

Within the initial set of freemium users, finding out the professors who have seen a significant change in student performance metrics and asking them to post testimonials about the product.

Pilot Program

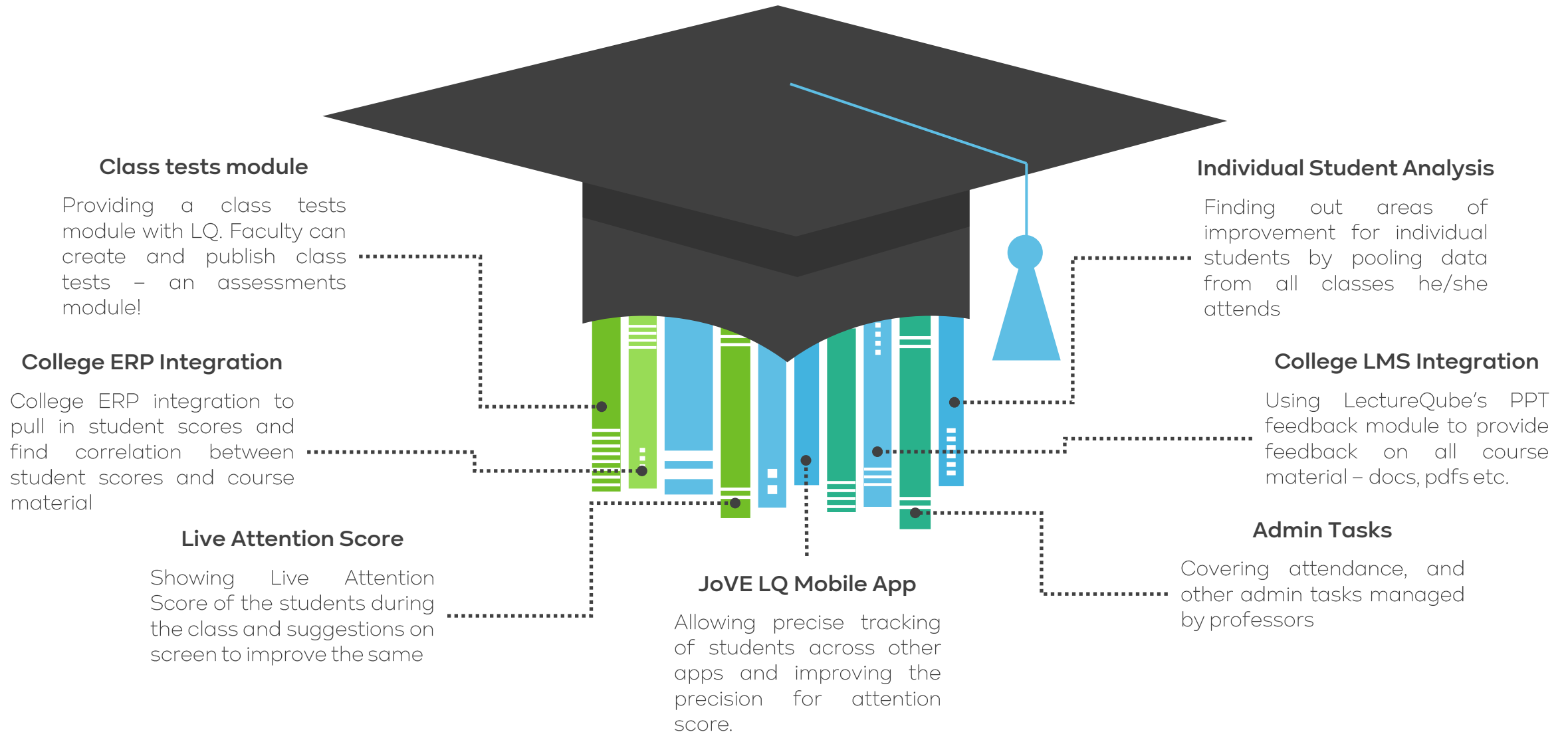
Allowing current users (professors and faculty) of JoVE to use JoVE LQ for free for 3-4 months (an average semester). Additionally, collaboration can be done on the college level as well.

Regular Product Webinars

Get a modern PowerPoint Presentation that is beautifully designed. Easy to change colors, photos and Text.



Steps ahead for JoVE LQ



THANK YOU!

ANY QUESTIONS?

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